



UDMnt

Installation Guide

May 2026

Document Revision: 4.20

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Revision History

Date	Revision	Description
May 2026	4.20	- Removed obsolete single axis breakout box accessory - Updated the motor types in Motor Drive
Sep 2025	4.10	Added a note about the PEG output frequency to Digital Inputs / Outputs
Jan 2025	4.00a	Updated Safety, EMC, and Wiring Guidelines
Dec 2024	4.00	Added reference to the PEG and Mark Application Note in Position Event Generation (PEG) Outputs
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Sep 2022	3.12.01	Add drive supply ripple note Correct digital I/O diagrams
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Aug 2016	2.25.20	Updated related documents

Date	Revision	Description
Jul 2016	2.25.10	Reformatted document Replaced connector image for J3, J4 connector Added a table describing the assignment of digital Inputs for MARK inputs Added "U" I/O configuration option to field #8 to the Configuration as Indicated by P/N table Added description of wiring color codes for five-phase motor Added reference to SETCONF for changing digital outputs from mechanical brake to general purpose. Various minor updates
Jul 2014	2.25	First Release

Conventions

We use the following conventions in this document.

Text Formats

Format	Description
Bold	Names of UI objects or commands
BOLD + UPPERCASE	ACSPL+ variables and commands
<code>Monospace + gray background</code>	Code example
<i>Italics</i>	Names of other documents
Blue	Hyperlink
[]	In commands indicates optional item(s)
	In commands indicates either/or items

Flagged Text

	Note - includes more information or programming tips.
	Caution - describes a condition that may result in damage to equipment.
	Warning - describes a condition that may result in serious bodily injury or death.

Related Documents

You can find related documentation on our website. The latest version of all documents can be downloaded by authorized users from [ACS Downloads](#).

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1. Introduction

1.1 Document Scope

This document describes the installation information for the UDMnt, including electrical interfacing, device compatibility, mounting, and ventilation.

2. UDMnt Overview

This chapter provides an overview of the UDMnt, the available product options as well as all the available kits and accessories associated with it.

2.1 Description

The UDMnt is a line of dual- and single-axis EtherCAT drives, compatible with any of ACS Motion Controller and EtherCAT master.

Figure 2-1 provides an overview of the UDMnt interface.

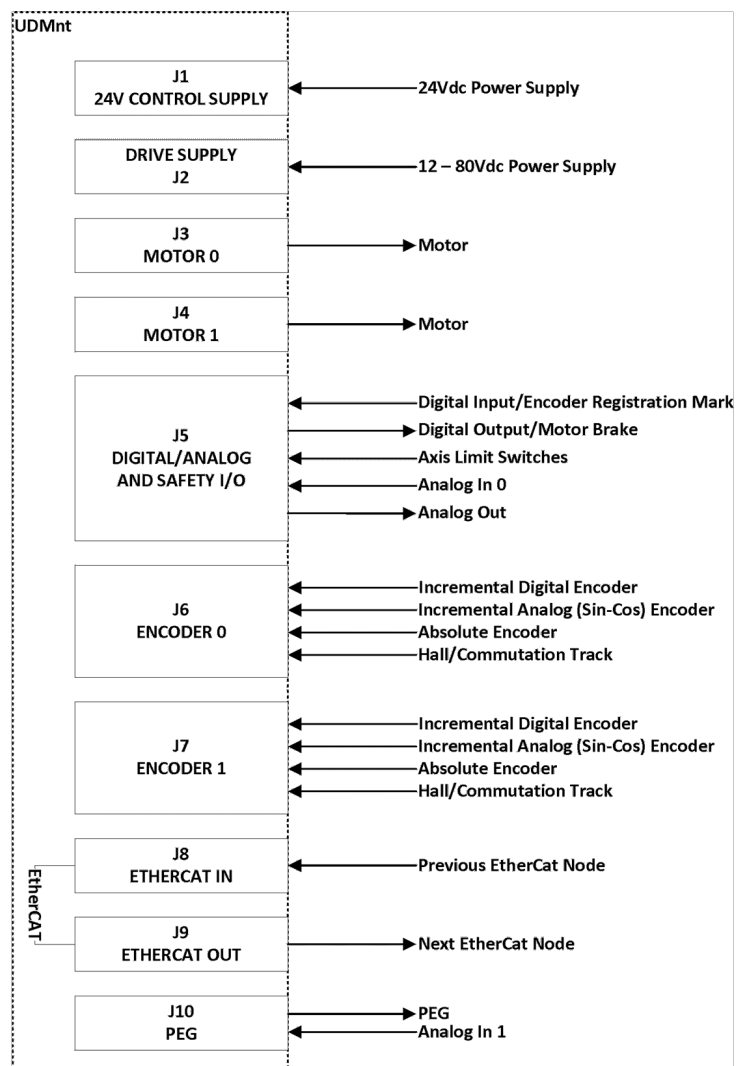


Figure 2-1. UDMnt Interface Block Diagram

2.2 Connectors

The figure and table below show the location and description of the connectors on the UDMnt unit.

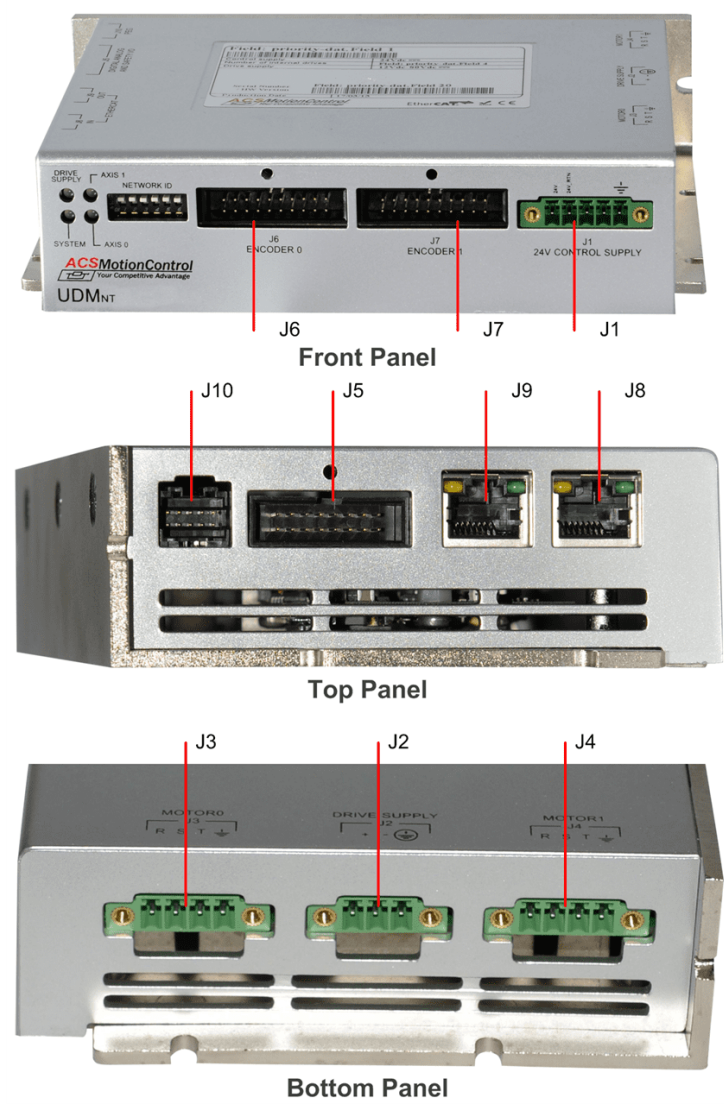


Figure 2-2. UDMnt Connectors - Location

Table 2-1. UDMnt Connectors

Connector	Name	Description
J1	24V CONTROL SUPPLY	Powers all low voltage / logic circuitry
J2	DRIVE SUPPLY	Powers the drives and motors
J3	MOTOR 0	Motor 0
J4	MOTOR 1	Motor 1

Connector	Name	Description
J5	DIGITAL/ANALOG and SAFETY I/O	General purpose inputs / encoder registration mark inputs, analog Input 0, axes' Limit Switch inputs
J6	ENCODER 0	Encoder and Hall inputs for MOTOR 0 (Axis 0)
J7	ENCODER 1	Encoder and Hall inputs for MOTOR 1 (Axis 1)
J8	ETHERCAT IN	EtherCat input
J9	ETHERCAT OUT	EtherCat output
J10	PEG	PEG, analog input 1



The Network ID dip switch is not used and its setting has no effect on the operation of the product.

2.3 Indicators

The following figures and tables show the location and description of the LED indicators on the UDMnt.



Figure 2-3. UDMnt Indicators - Front Panel

Table 2-2. UDMnt Front Panel Indicators - Description

LED	Description	Notes
Axis 0 Axis 1	Dual color LED: > Green - drive is on > Red - drive is off due to fault > Off - drive is off	
Drive Supply	Green LED: > On - drive supply is on > Off - drive supply is off	

LED	Description	Notes
Systems	Dual color LED: > Red/Off - communication fault > Green/Blinking - normal operation - master communicating with slave	See the <i>EtherCAT Network Diagnostics</i> for troubleshooting.

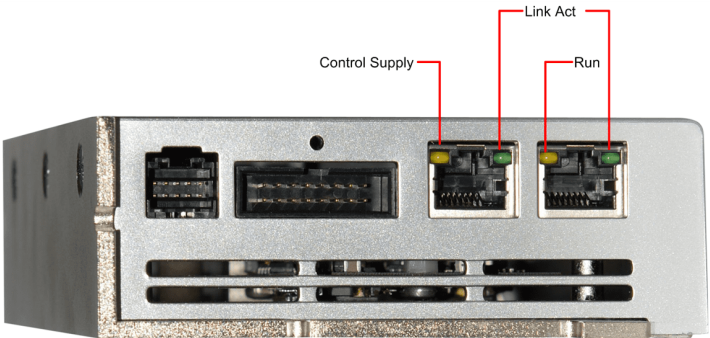


Figure 2-4. UDMnt Indicators - Top Panel

Table 2-3. UDMnt Top Panel Indicators

LED	Description	Notes
Link Act	Green LED: > On - connected but not communicating with master > Blinking - communicating with master > Off - not connected	One on each EtherCAT port.
Run	Yellow LED: > On - network communication is OK > Blinking/Off - network communication error	
Control Supply	Yellow LED: > On - control supply is on > Off - control supply is off	

2.4 Order Part Number

The ordering part number (P/N) contains eight characters that designate the configuration ordered for the UDMnt module as explained in [Table 2-4](#).

Table 2-4. Configuration as indicated by P/N

Ordering Options	Field	Example	Values
Number of axes	1	2	1, 2
Continuous Current (Peak is double)	2	B	A = 2.5/5A B = 5/10A C = 10/20A
Total number of encoder channels	3	2	1 (for single-axis unit only), 2
500kHz SinCos	4	0	0, 1, 2
10MHz SinCos	5	1	0, 1, 2
Absolute encoders type	6	N	U = User selectable N = None E = EnDAT 2.1(digital)/2.2 S = SmartAbs P = Panasonic B = BiSS-C I = SSI
Number of absolute encoders interfaces	7	0	0, 1, 2
I/O configuration	8	N	N - Inputs & limits: 24 V / SOURCE (PNP), Outputs: 24 V / SOURCE (PNP). D - Identical to (N), For compatability reasons. S - Inputs & limits: 24 V / SINK (NPN). Outputs: 24 V / SOURCE (PNP). R - Inputs & limits: 5 V / SOURCE (PNP). Outputs: 5 V / SOURCE (PNP). T - Inputs & limits: 5 V / SINK (NPN). Outputs: 5 V / SOURCE (PNP). U- Outputs & Inputs: 24 V / SOURCE (PNP), Limits: 24 V / SINK (NPN) ²

¹All absolute encoder channels must be the same type

²Sink/Source option requires external power supply. No Internal supply is available.

For example, P/N UDMnt2B200N0D, represents the configuration described in [Table 2-5](#).

Table 2-5. P/N Example

Field		1	2	3	4	5	6	7	8
P/N	UDMnt	2	B	2	0	0	N	0	D

TheUDMnt is shipped with the configuration set as ordered. The configuration cannot be modified by the user.

2.5 UDMnt Package Content

The UDMnt package contains the following items:

- > UDMnt Module
- > Control Supply Mating Connector and Locking Clips Kit

2.5.1 Control Supply Mating Connector and Locking Clips Kit

This kit comes with the UDMnt.

ACS P/N: CK-169000-000/LF.

Table 2-6. Control Supply Mating Connector and Locking Clips Content

Ref.	P/N	Qty	Manufacturer	Description
1	1827732	1	Phoenix	24V CONTROL SUPPLY mating connector (J1) MC1,5/3-STF-3.81
2	GG-16900-00	2	ACS	Locking clip for encoder mating connectors (J6, J7) (see note below)
3	GG-16900-01	1	ACS	Locking clip for Digital, ANalog and Limit I/O mating connector (J5)
4	SC-10036-120	3	Many	Phillips type screw 4 x 3/8



The two types of clips are similar. Make sure to use the proper one for each connector. The difference between the two types is shown in [Figure 2-5](#) and [Figure 2-6](#).

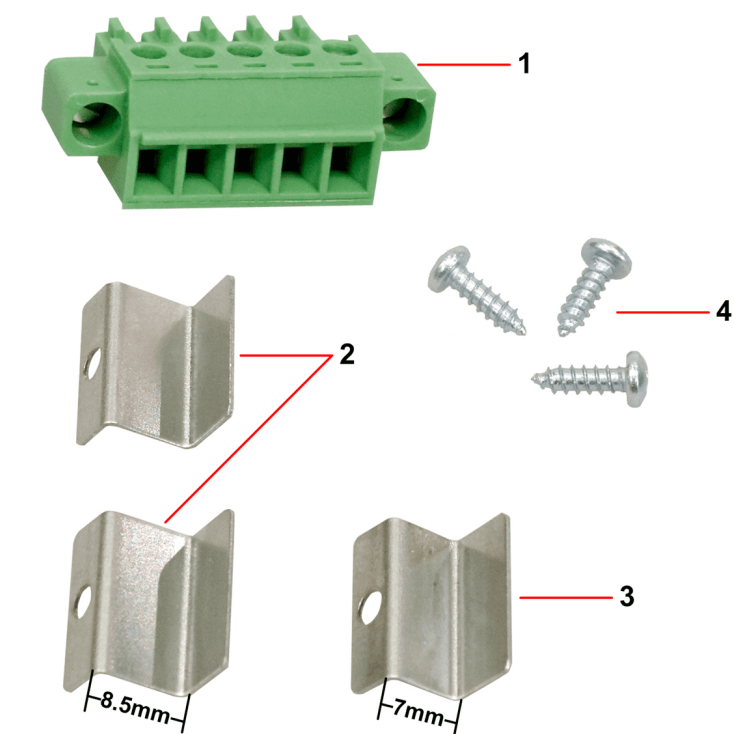


Figure 2-5. Control Supply Mating Connector and Locking Clips Kit

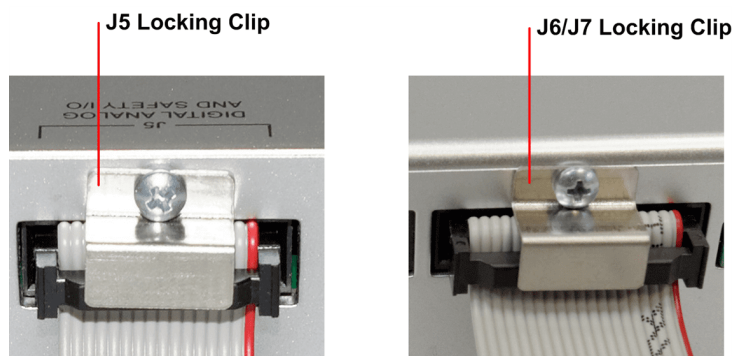


Figure 2-6. Connector Locking Clips

2.6 Optional Kits and Accessories

The following kits are available:

- > **UDMNT MATING CONNECTORS AND ACCESSORIES KIT** - A set of mating connectors, excluding the 24V CONTROL SUPPLY (J1) mating connector that is included in the UDMnt package.
- > **UDMNT DUAL AXIS BREAKOUT BOX ACCESSORY KIT** - A set of cables, connectors and breakout modules for the dual-axis UDMnt. Recommended for fast prototyping.

2.6.1 UDMnt Mating Connectors and Accessories Kit

ACS P/N: UDMnt-ACC1.

Table 2-7. UDMnt Mating Connectors and Accessories Kit Content

Ref.	Manufacturer	Manufacturer P/N	Part Description	Qty
1	Phoenix	1827716	DRIVE SUPPLY (J2) mating connector MC 1.5/3-STF-3.81	1
2	Phoenix	1827729	MOTOR (J3,J4) mating connectors MC 1.5/4-STF-3.81	2
	DEGSON	15EDGKM-3.81- 04P-14-00AH		
3	3M	89120-0101HA	Encoder mating connectors (J6,J7) Header 20- pin, female 2.54mm	2
	3M	3448-89120	Strain Relief Clip	2
4	3M	89116-0101HA	Digital, Analog and Safety I/O (J5) mating connector Header 16-pin, female 2.54mm	1
	3M	3448-89116	Strain Relief Clip	1
5	JST	PADP-10V-1-S	PEG mating connector housing (J10)	1
6	JST	SPND-001T-C0.5	Mating Connector pins	10

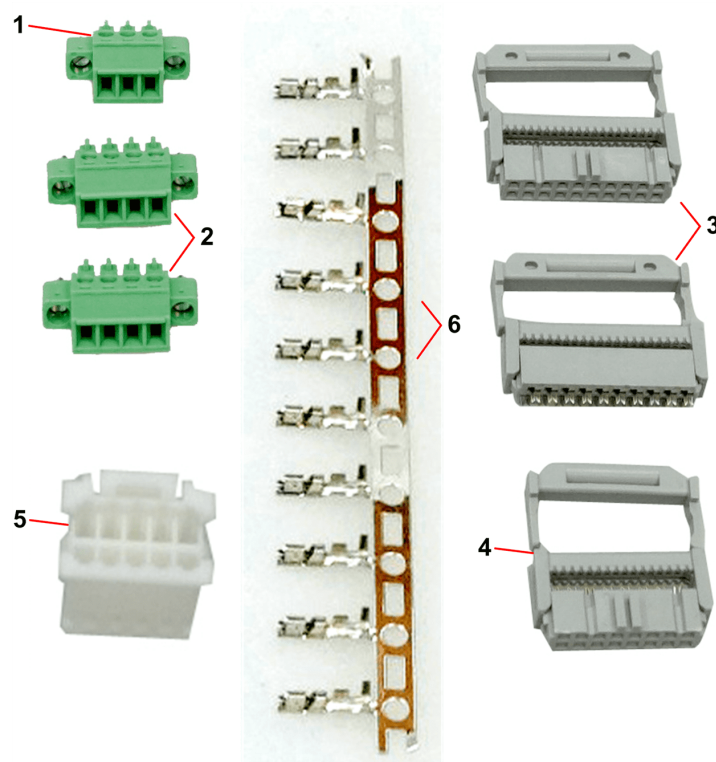


Figure 2-7. UDMnt Mating Connectors and Accessories Kit

2.6.2 UDMnt Dual Axis Breakout Box Accessory Kit

ACS P/N: UDMnt-2-BOB.

Table 2-8. UDMnt Dual Axis Breakout Box Accessory Kit

Ref.	P/N	Qty.	Manufacturer	Part Description
1	CB-16900-000/LF	1	ACS	0.5m cable with mating connector to PEG (J10)
2	1827716	1	Phoenix	DRIVE SUPPLY (J2) mating connector MC 1.5/3-STF-3.81
3	1827729	2	Phoenix	MOTOR (J3,J4) mating connectors MC 1.5/4-STF-3.81
4	2962641	1	Phoenix	Digital, Analog and Safety I/O (J5) UM45-FLKS16 VARIOFACE module. 16 positions
5	2962654	21	Phoenix	Encoder(J6, J7) UM45-FLKS20 VARIOFACE module. 20 positions
6	FC-01616-50	1	ACS	0.5m flat ribbon cable,

Ref.	P/N	Qty.	Manufacturer	Part Description
				16-pin
7	FC-02020-050	1	ACS	0.5m flat ribbon cable, 20-pin

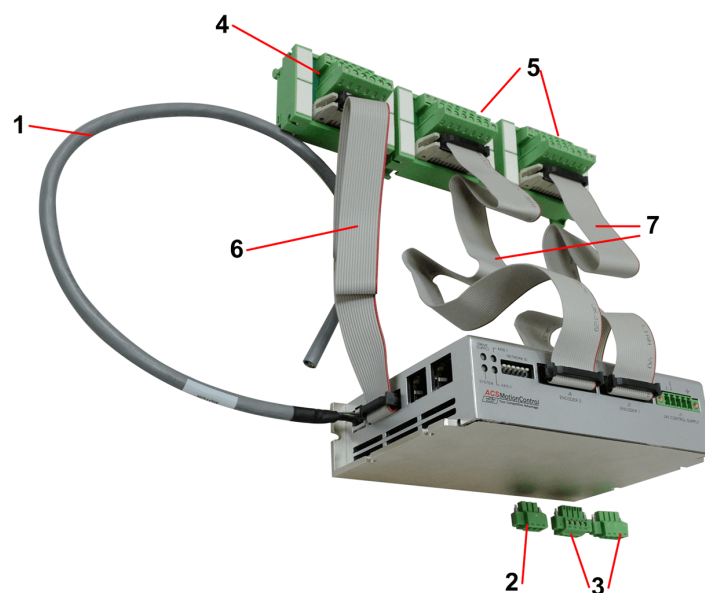


Figure 2-8. UDMnt Dual Axis Breakout Box Accessory Kit

2.6.3 EtherCAT Cables

The following Ethernet CAT5e cables are offered by ACS:

Table 2-9. EtherCAT cables offered by ACS

Ethernet Cable, CAT5 Length	
30cm	SP+ECAT-CA-30CM-00
0.5m	SP+ECAT-CA-50CM-00
1.0m	SP+ECAT-CA-1M-00
5.0m	SP+ECAT-CA-5M-00
10.0m	SP+ECAT-CA-10M-00
20.0m	SP+ECAT-CA-20M-00

2.7 Additional Accessories Needed

The UDMnt requires two power supplies to be provided by the user:

- > CONTROL SUPPLY – A 24 VDC power supply. Isolated and UL certified. Rated for >1A. This power supply is connected to J1 (24V CONTROL SUPPLY)

- > DRIVE SUPPLY – A 12 VDC to 80 VDC power supply. Isolated and UL certified. Its current should be rated so that it can provide the total RMS current of the motors as well as the short time peak current. This power supply is connected to J2 (DRIVE SUPPLY).

The UDMnt has no regeneration circuit. To ensure that the drive voltage does not exceed the 83V maximum voltage during deceleration, a regeneration device, capable of absorbing the motors' regenerating power, might be needed.

The drive will be disabled when the voltage reaches 85V. If the voltage rises above 100V the UDMnt might be damaged.

To comply with European standards (CE), we recommend that you use an AC line filter – such as Shaffner FN-2060-16-06, or similar, on the AC input lines that supply the AC/DC power supply.

3. Mounting and Cooling

The UDMnt should be mounted vertically, using M4 type Philips screws. The dimensions (in millimeters) for the UDMnt mounting are shown in [Figure 3-1](#).

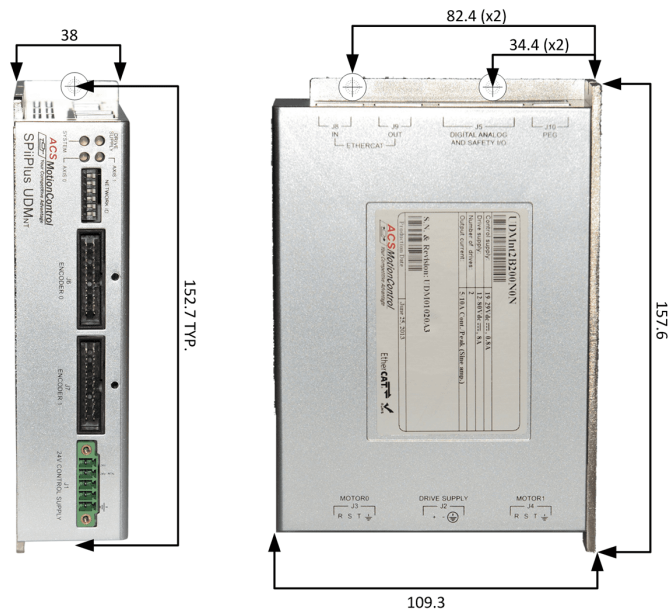


Figure 3-1. UDMnt Vertical Mounting Options

Leave sufficient clearance on all open sides for cable routing and free air flow as shown in [Figure 3-2](#).

The UDMnt is designed to operate at full power with natural air convection, without any forced air for cooling at an ambient temperature (the temperature of its surroundings) of up to 50°C.

The maximum temperature raise of the heat sink under full load, when the product is mounted vertically, is 10°C.

We recommend that you do not mount the unit horizontally. However, if the unit is mounted horizontally, mount it with the metal heat sink on top and provide forced horizontal airflow of at least 10CFM to cool the internal circuits.

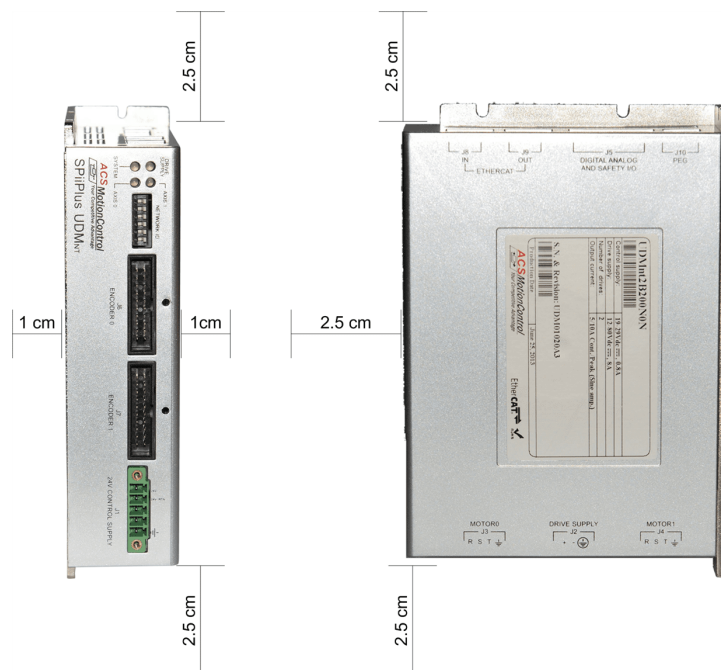


Figure 3-2. UDMnt Vertical Mounting Airflow

4. Connections

This chapter describes in details how to interface with the UDMnt following proper safety, EMC, and wiring guidelines.

4.1 Safety, EMC, and Wiring Guidelines



WARNING Read this section carefully before beginning the installation process.

Make sure that the following guidelines and procedures are addressed and observed prior to powering up and while handling any of the EtherCAT network elements.

Installation and maintenance must be performed only by qualified personnel who have been trained and certified to install and maintain lasers, high power electrical and electromechanical equipment, servo systems, power conversion equipment, and distributed networks.

Prior to powering up the system, ensure that all EtherCAT network devices are properly installed and grounded.

Ensure that all the attached power and signal cables are in good operating condition. Only perform maintenance after the relevant network devices have been powered down and all associated and surrounding moving parts have settled in their safe mode of operation. Certain drives require a longer time to fully discharge.

To avoid electrical arcing and hazards to personnel and electrical contacts, avoid connecting and disconnecting the product while the power source is on. Do not connect or disconnect a motor from a drive when the drive is enabled or in any condition in which there is current through the motor phases.

When connecting the product to an approved isolated control and drive supply, connect it through a line that is separated from hazardous live voltages using reinforced or double insulation, in accordance with approved safety standards.



WARNING The product is not intended for use in safety-critical applications (such as life support devices) where a failure of the product can reasonably be expected to cause severe personal injury or death.

Perform the following instructions to ensure safe and proper wiring:

- Whenever possible, use shielded cables with braided shield of at least 80%-95% coverage.
- Proper wiring, grounding, and shielding are essential for ensuring safe, dependable, and optimal servo performance.
- After completing the wiring, carefully inspect all wires to ensure tightness, good solder joints, and general safety.

Follow the guidance of [Table 4-1](#), based on the current rating of your product.

Table 4-1. Wiring Guidelines

	Gauge	Twisted Pair
Control power supply	18AWG	No
Drive power supply	16AWG	No
Motor 2.5/5 (sine-peak/max, A)	22AWG or 2X24AWG	No
Motor 5/10 (sine-peak/max, A)	20AWG	No
Motor 10/20 (sine-peak/max, A)	16AWG	No
Motor brake	18AWG	No
Encoders	28AWG (up to 0.6A) 28AWG (up to 1A)	Yes
Digital I/O	28AWG	According to application demands
Analog I/O	28AWG	Yes

4.1.1 UL Installation

- These products were not evaluated for Overload protection or internal speed-sensitive overload protection, thermal memory retention or provisions to receive and act upon signals from remote devices for motor overtemperature protection. If such protection is needed in the end-use product, it needs to be provided by additional means.
- The product is to be used with cord connected end products or if not cord connected the end product should be provided with suitable UL 1449 Overvoltage protection so that the rated impulse voltage rating(Vp) is no larger than 0.8 kV, type 2 per Table 8.1/UL 840.
- This device is to be used within a controlled environment, Pollution Degree 2.
- There is no isolation between power circuits and low voltage signal circuits(Encoders, Hall Inputs, Analog Input/Outputs). Isolation should be considered in the end product.



WARNING J3 and J4 contain hazardous voltages of 80V PWM modulated.

4.2 Connecting the UDMnt

Connect the UDMnt as described below:



In some cases, not all steps listed below apply (for example, when connecting to a single axis UDMnt, only Motor 0 connections are relevant).

1. Ensure that all power supplies are off when preparing the unit.
2. Connect 24 VDC control supply to J1.

3. Connect 12 to 80 VDC drive supply to J2.
4. Connect motor 0 to J3.
5. Connect motor 1 to J4.
6. Connect the feedback sensors (ENCODER0 for Motor 0) to J6 and (ENCODER1 for Motor 1) to J7.
7. Connect I/Os to J5 and J10.
8. Connect the EtherCAT IN cable (from the EtherCAT Master or another EtherCAT slave) to J8.
9. Connecting J9 (EtherCAT OUT)
 - a. When the UDMnt is not the last network node, connect the EtherCAT cable between J9 and EtherCAT IN of the next EtherCAT slave.
 - b. When the UDMnt is the last network node and a ring topology is used, connect J9 to the EtherCAT Master secondary port.
 - c. When the UDMnt is the last network node and a line topology is used, leave J9 unconnected.
10. Once the unit is connected:
 - a. Turn on the 24 VDC control supply and verify communication with the UDMnt.
 - b. Turn on the 12 to 80 VDC drive supply.



The supplies can be turned on and off in any order.

4.3 Power Supplies

The UDMnt is fed by two power supplies:

- > Control Supply - 24 VDC (J1)
- > Drive Supply – 12 to 80 VDC (J2)

The power supplies must be provided by the customer and be UL certified. The supplies can be switched on in any order.

During emergency situations, the Drive Supply can be disconnected while the Control Supply should remain connected.

Each power supply has a LED indicator on the UDMnt.

4.3.1 24V Control Supply

An external 24 VDC isolated power supply (not included with the UDMnt) feeds all logic and control low voltage circuitry.

We recommend that you keep this power supply active (on) also during emergency stop situations, thus ensuring the continuing operation of the network, the controller, the feedback sensors, and I/Os.

4.3.1.1 24 VDC Power Supply Specifications

The 24 VDC power supply should comply with the specifications shown in [Table 4-2](#).

Table 4-2. 24 VDC Power Supply Requirements

	Description
Type	Isolated Low noise UL certified
Output voltage range	24 VDC±20%
Output current	Minimum 1A

Example: Lambda, P/N LS100-24

4.3.1.2 J1 - 24 VDC Control Supply Connector

- > Label: J1 24V CONTROL SUPPLY
- > Connector: MC 1.5/ 5-GF-3.81, by Phoenix, PN 1827897
- > Mating connector: MC 1.5/ 5-STF-3.81, by Phoenix, PN 1827732

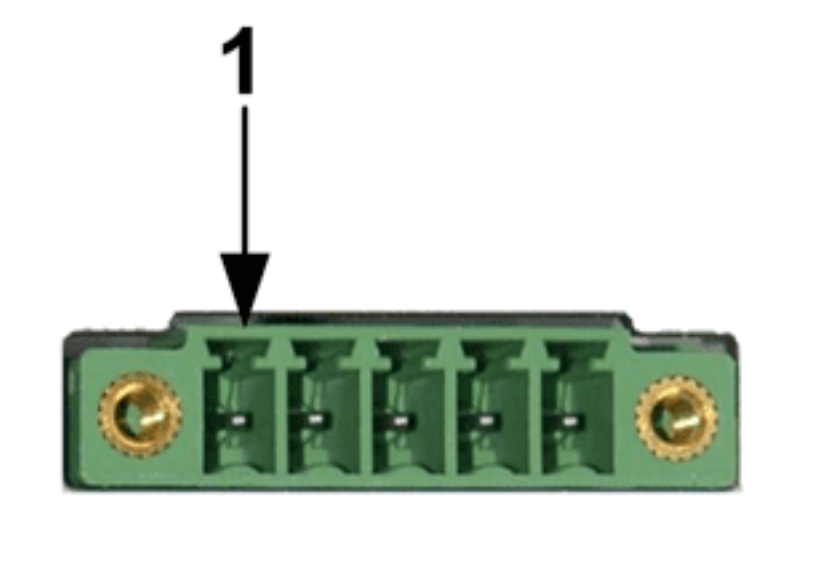


Figure 4-1. J1 -24V Control Supply Connector

Table 4-3. 24 V Control Supply Pinout

Pin	Signal	Description
1	24 VDC	+24 VDC
2	24VRTN	+24V dc return
3	NC	Not connected
4	NC	Not connected
5	Shield	Electrical ground

4.3.1.3 Connection Instructions

Use a shielded cable with a minimum gauge of 18 AWG.

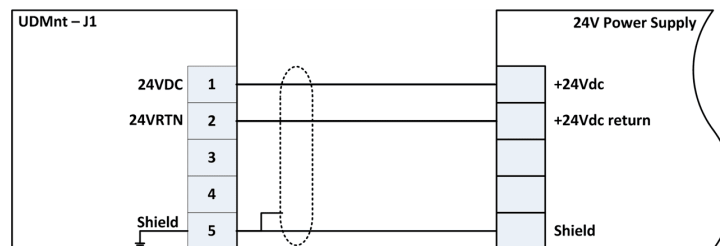


Figure 4-2. 24V Control Supply Connections

4.3.2 Drive Supply

An external isolated 12 VDC to 80 VDC power supply (not included with the UDMnt) feeds the drives and the motors. The maximum drive supply is not to exceed 80 VDC.

4.3.2.1 12V to 80V Drive Supply Requirements

This power supply should comply with the following specifications.

Table 4-4. 12V to 80V Drive Supply Requirements

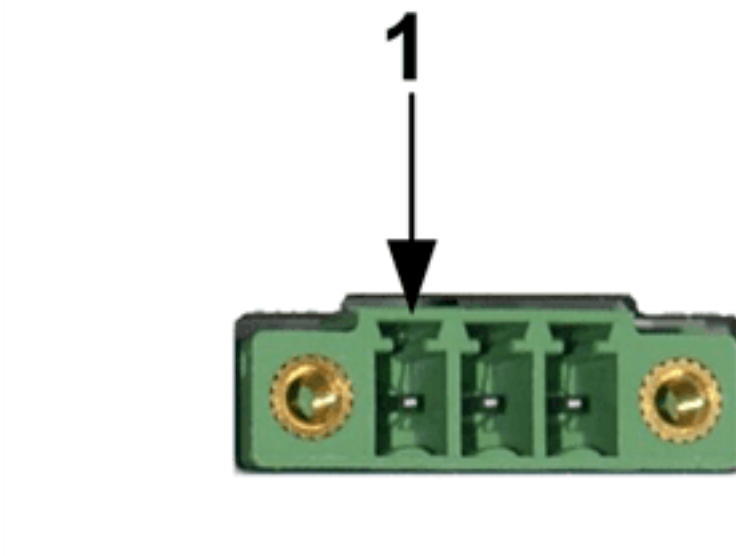
	Description	Remarks
Type	Isolated	
Output voltage range	12 VDC to 80 VDC	Output voltage not to exceed 83 VDC
Output current, maximum	8A	The total input current of the UDMnt is limited to 8A



European standards (CE) require the use of an AC line filter on the AC input lines that supply the 24 VDC control supply.

4.3.2.2 J2 - Drive Supply Connector

- > Label: J2 DRIVE SUPPLY
- > Mating connector: Phoenix, PN 1827716

**Figure 4-3. J2 - Drive Supply Connector****Table 4-5. J2 - Drive Supply Connector Pinout**

Pin	Signal	Description
1	VP+	Drive Supply +
2	VP-	Drive Supply -
3	PE	Protective Earth

4.3.2.3 Connection Instructions

Use a cable with a minimum gauge of 16 AWG.

Route the drive supply and motor cables as far as possible from all other noise-sensitive cables (such as encoders and I/O).

1. Connect a 10A fuse between the UDMnt and the external drive supply.



We recommend a fast-acting fuse, rated current 10A. Example: Cooper Bussmann MNF P/N KTK-R-10.

2. The UDMnt has no regeneration circuit.
 - a. If the drive voltage rises due to regeneration, we recommend that you use a drive supply with a maximum of 75 VDC.
 - b. To keep the drive supply in range, use an active regeneration device. We recommend a shunt regulator (regeneration circuit) set to $80V \pm 3\%$.



The drive supply must not exceed 83 VDC. It can be disconnected during E-Stop safety situations to prevent any motion of the axes.

3. Connect the UDMnt PE (Protective Earth) to the drive supply PE.

4. In noisy environments, for better noise immunity, the user may consider connecting an external PE (Protective Earth) to the VP-point (J2 pin 2) as well as to J2 pin 3. Connect a short-cable between J2 pin 3 and J2 pin 2.



Connecting the J2 pin 3 to J2 pin 2 disrupts the isolation between the motor circuits and the PE (Protective Earth), and thus impacts adherence to safety standards and regulation.

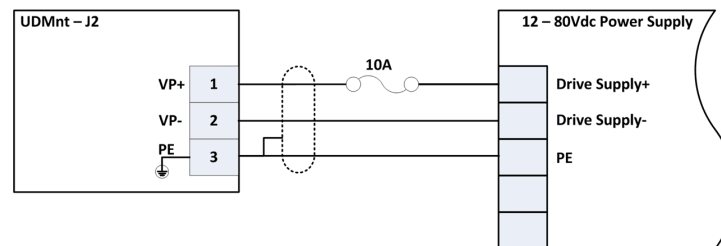


Figure 4-4. Drive Supply Connections



Measure the ripple next to the drive supply connector and verify that the voltage ripple is not more than 1 V during motion. If the ripple is higher, add external capacitors close to the drive to reduce the ripple.

4.4 Motors

The UDMnt supports the following rotary and linear motors:

- > Single-phase (DC brush, moving coil), see [Figure 4-6](#).
- > Two-phase (AC synchronous, step motor), see [Figure 4-7](#).
- > Three-phase (AC synchronous, step motor), see [Figure 4-8](#).
- > Five-phase (step motor. Consult company), see [Figure 4-9](#).



For more information about each motor type, see *SPiiPlus MMI Application Studio User Guide* and applicable application notes.



Five-phase step motor is an order option and is not supported by the standard unit.

4.4.1 J3 (MOTOR0), J4 (MOTOR1) Connectors



WARNING J3 and J4 contain hazardous voltages of 80V PWM modulated.

- > Label: J3 MOTOR0, J4 MOTOR1
- > Mating connector: MC 1.5/ 4-STF-3.81, by Phoenix, PN 1827729

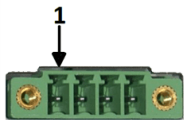


Figure 4-5. J3, J4 - Motor Connectors

Table 4-6. J3, J4 - Motor Connectors Pinout

Pin	Signal	Description
1	R	Drive Phase R
2	S	Drive Phase S
3	T	Drive Phase T
4	Shield	Electrical ground / Protective earth

4.4.2 Connection Instructions

Use a shielded cable with a minimum gauge of 16 AWG. It should be less than 10 meters long.

1. Connect the motors to J3 and/or J4 according to Figure 4-6 through Figure 4-9.



Only one five-phase step motor can be used, connected to both J3 and J4 connectors.

2. Route the motors' cable (and the drive supply cable) as far as possible from all other noise-sensitive cables (such as encoders and I/O).

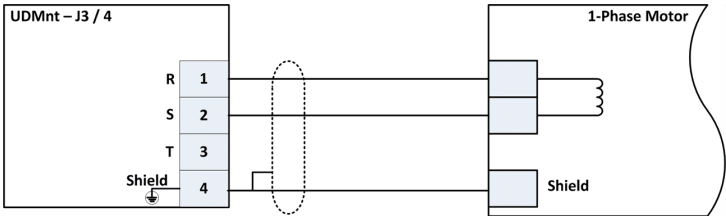


Figure 4-6. Single-Phase Motors (DC brush, Voice coil) Connections

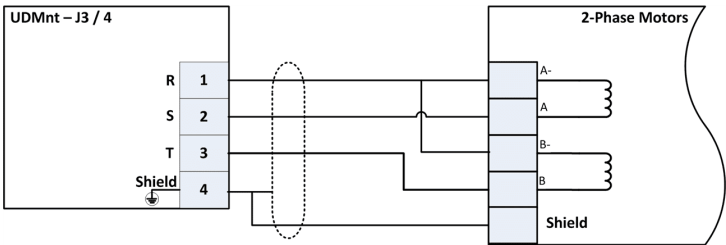


Figure 4-7. Two-Phase Motors (AC synchronous, Step Motor)

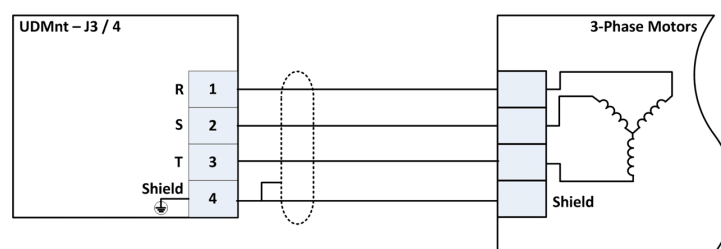


Figure 4-8. Three-Phase Motors (AC synchronous, Step Motor)

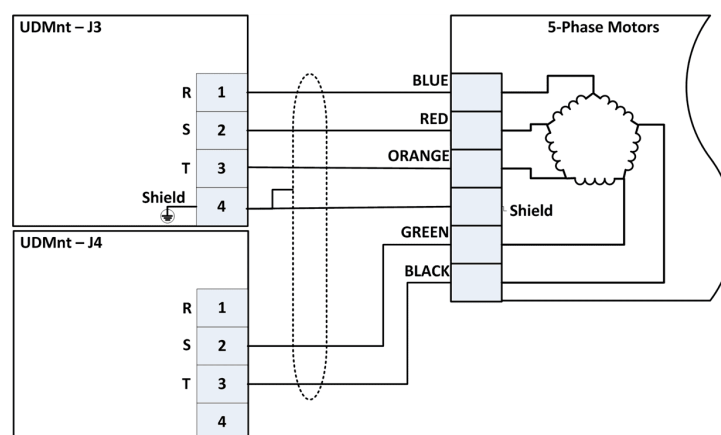


Figure 4-9. Five-Phase Step Motor Connections (example of an Oriental Motor)

Table 4-7. Five-Phase Motor Wiring Color Codes (example of Oriental Motor)

Motor Phase	Color
Phase R of AXIS 0	Blue
Phase S of axis 0	Red
Phase T of axis 0	Orange
Phase S of axis 1	Green
Phase T of axis 1	Black

4.5 Feedback Sensors

There are two position feedback sensors interfaces, supporting the following types of sensors:

- > AqB,I and Clk/Dir (=Pulse/Dir) incremental digital encoders, with or without Hall sensors or commutation tracks.
- > AqB,I incremental analog encoder (SinCos) with or without Hall sensors or commutation tracks. There are two types of interfaces:
 - > 500kHz SinCos. The speed is limited to 500kHz frequency. This interface should be used with optical SinCos encoders.
 - > 5MHz SinCos. The speed is limited to 5MHz frequency. This interface should be used with Laser Encoders for higher resolution while moving at high speeds.
- > Absolute serial encoders of the following types:

- > Data/Clock
 - > EnDAT 2.2
 - > BiSS-B/C (SSI)
- > Data Line Only (non-return-to-zero NRZ)
 - > Tamagawa Smart-Abs
 - > Panasonic

Encoder and Hall signals share the same connector.

Connect the feedback sensors for Motor 0 to J6 and for Motor 1 to J7.

When using dual feedback (for Motor 0 only), connect the primary feedback sensors to J6 and the secondary feedback sensors to J7.



Interfaces for SinCos and for absolute encoders is optional and must be specified when ordering the product.



UDMnt does not support single-ended encoders.

All feedback sensors are powered by the UDMnt internal 5.1 VDC power supply. The total consumptions of all sensors (encoders and Hall) should not exceed 0.5A.

Make sure that the actual voltage measured on the sensors contacts is more than its minimum specification (typically >4.75 VDC).

4.5.1 J6, J7 (Encoder 0, Encoder 1) Feedback Sensor Connectors



stands for either 0 (J6) or 1 (J7).

- > Labels: J6 ENCODER 0, J7 ENCODER 1
- > Mating connector: Header 20-pin, female 2.54mm. M3, PN 89120-0101HA and 3448-89120

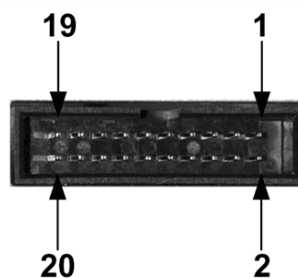


Figure 4-10. J6, J7 - Feedback Sensors Connectors

Pin	Signal	Incremental Digital Encoder AqB,I	Incremental Digital Encoder Clk/Dir,I	Incremental Analog Encoder	Absolute Encoder DATA/CLOCK	Absolute Encoder DATA Only
1	5U	+5V supply	+5V supply	Not used	+5V supply	+5V supply
2	5U_RTN	+5V return	+5V return	Not used	+5V return	+5V return
3	CHA#-	Channel A-	Clk-	Not used	Data-	Data-
4	CHA#+	Channel A+	Clk+	Not used	Data+	Data+
5	CHB#-	Channel B-	Dir-	Not used	Clock-	Not used
6	CHB#+	Channel B+	Dir+	Not used	Clock+	Not used
7	CHI#-	Index -	Index -	Not used	Not used	Not used
8	CHI#+	Index+	Index+	Not used	Not used	Not used
9	HA#	Hall sensor A	Hall sensor A	Hall sensor A	Hall sensor A	Hall sensor A
10	HB#	Hall sensor B	Hall sensor B	Hall sensor B	Hall sensor B	Hall sensor B
11	HC#	Hall sensor C	Hall sensor C	Hall sensor C	Hall sensor C	Hall sensor C
12	Shield	Shield	Shield	Shield	Shield	Shield
13	5F	Not used	Not used	+5V analog supply	Not used	Not used

Pin	Signal	Incremental Digital Encoder AqB,I	Incremental Digital Encoder Clk/Dir,I	Incremental Analog Encoder	Absolute Encoder DATA/CLOCK	Absolute Encoder DATA Only
14	5F_RTN	Not used	Not used	+5V analog return	Not used	Not used
15	SIN#-	Not used	Not used	Channel SIN-	Not used	Not used
16	SIN#+	Not used	Not used	Channel SIN+	Not used	Not used
17	COS#-	Not used	Not used	Channel COS-	Not used	Not used
18	COS#+	Not used	Not used	Channel COS+	Not used	Not used
19	SC_I#-	Not used	Not used	Index- (analog encoder)	Not used	Not used
20	SC_I#+ N	Not used	Not used	Index+ (analog encoder)	Not used	Not used



Hall sensors / commutation tracks are used for initial AC Synchronous motor commutation upon power up. Use of Hall sensors / commutation tracks for commutation purposes is optional. The controller can perform sinusoidal commutation also without Hall sensors / commutation tracks.

4.5.2 J6, J7 Connection Instructions

Use shielded cables with twisted pairs, a minimum gauge of 24 AWG and up to 20 meters in length.

1. Connect the motors to J3 J4 according to [Figure 4-11](#) through [Figure 4-15](#).
2. The mating connector can be locked using the provided connector clip and screw, see [Control Supply Mating Connector and Locking Clips Kit](#).

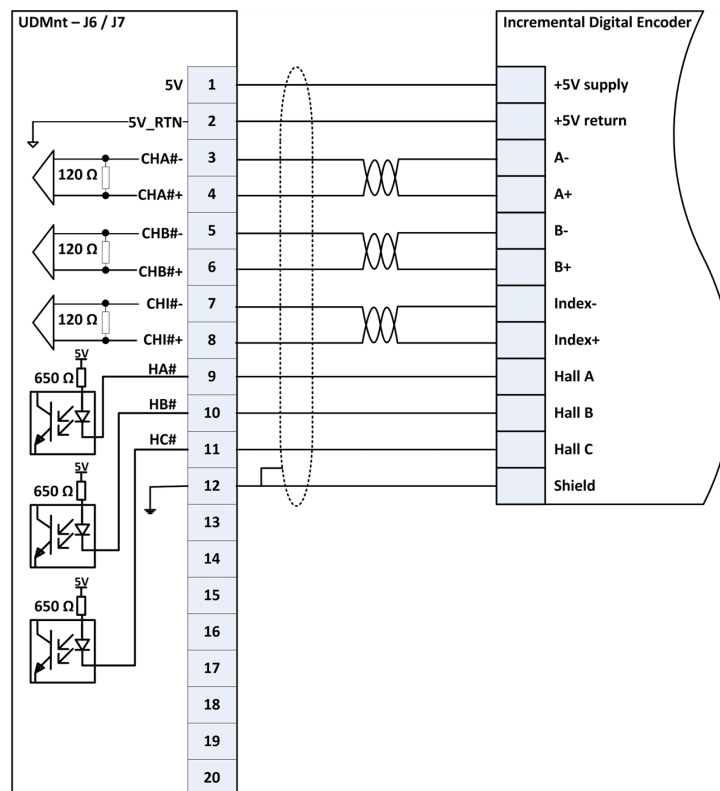


Figure 4-11. Incremental Digital Encoder with Hall / Commutation Tracks AqB,I Connections

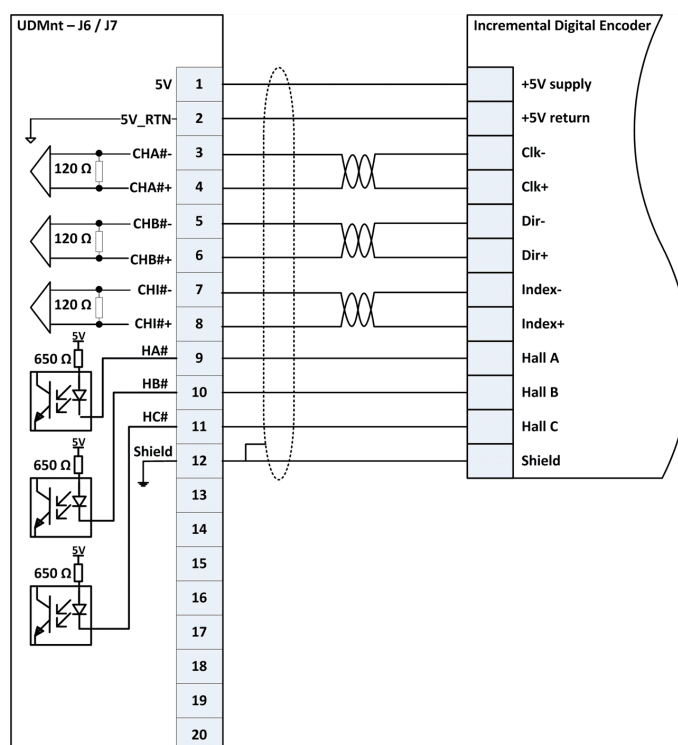


Figure 4-12. Incremental Digital Encoder with Hall / Commutation Tracks Clk/Dir, I Connections

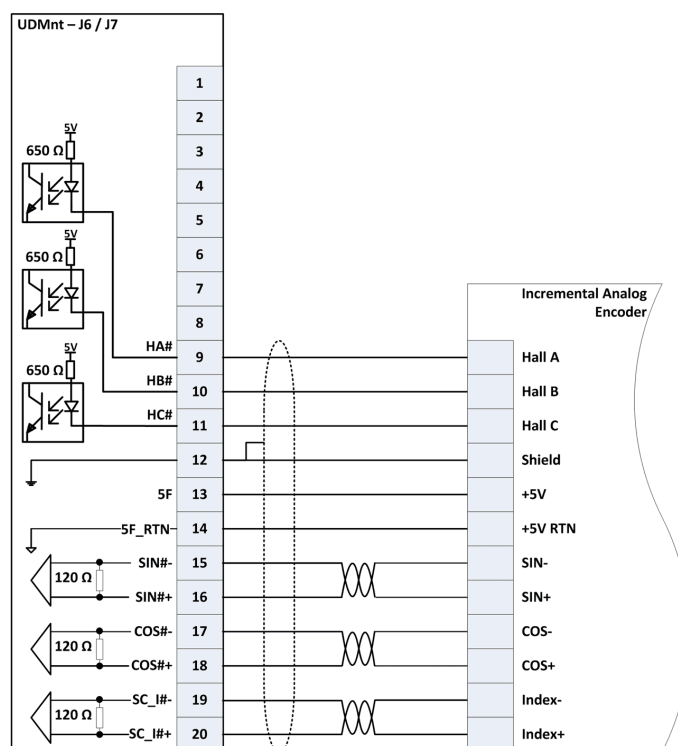


Figure 4-13. Incremental Analog Encoder (SinCos) with Hall / Commutation Tracks AqB,I Connections

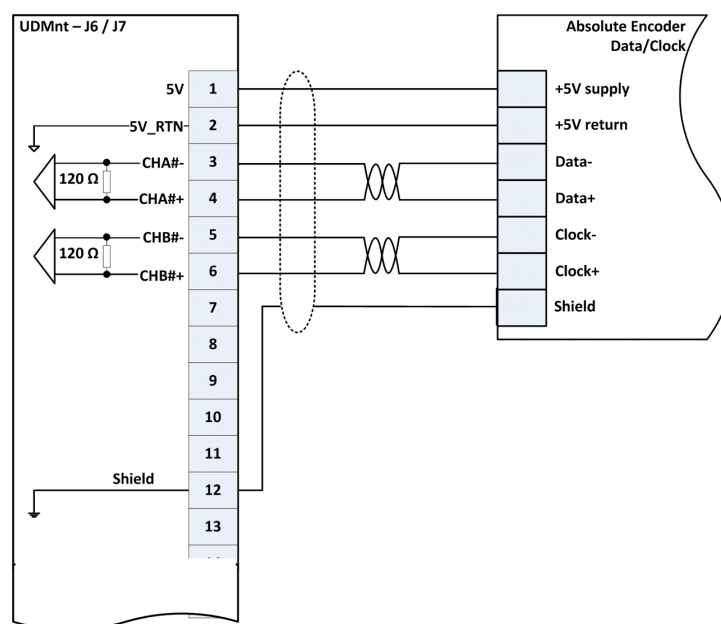


Figure 4-14. Absolute Serial Encoders with Data/Clock Connections (EnDAT 2.2, BiSS-B/C(SSI))

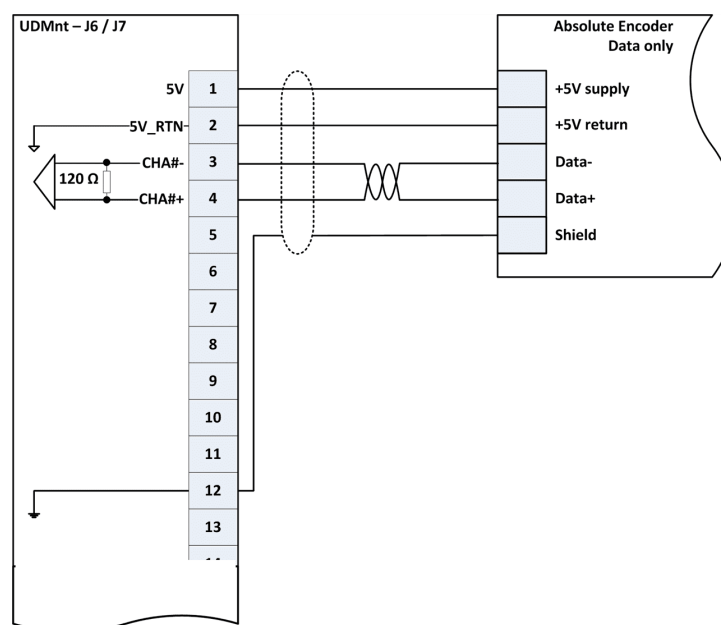


Figure 4-15. Absolute Serial Encoders with Data Line Only Connections (Tamagawa Smart-Abs, Panasonic)

4.6 Digital & Analog I/O

The UDMnt includes the following inputs/outputs:

Table 4-8. Digital & Analog I/O

I/O	Signal	Connector
Limit switch inputs	RLO, LLO, RL1, LL1	J5
Digital inputs / Encoder registration mark	IN0 – IN3	J5
Digital outputs	OUT0, OUT1	J5
Analog inputs	AIN0 AIN1	J5 J10
Analog output	AOUT	J5
PEG outputs	PEGO, PEG1	J10



With the UDMnt installed in a network, the user can obtain the exact mapping of the ACSPL+ IN, OUT, AIN and AOUT variables to the connector pins for the device in the system through the MMI Application Studio. For detailed instructions, see the *eSPiiPlus MMI Application Studio User Guide* under system configuration.

4.6.1 J5 - Digital, Analog and Safety I/O Connector

- > Label: J5 DIGITAL/ANALOG AND SAFETY I/O
- > Mating connector: Header 16-pin, female, 2.54mm. 3M, PN 89116-0101HA and 3448-89116

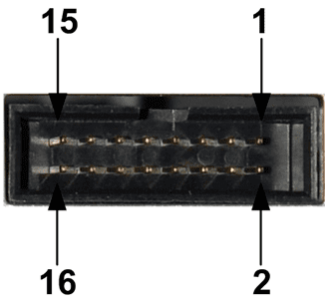


Figure 4-16. Digital/Analog and Safety I/O Connector

Table 4-9. Digital/Analog and Safety I/O Connector

Pin	Signal	Description
1	IN0	Digital input 0
2	IN1	Digital input 1
3	IN2	Digital input 2

Pin	Signal	Description
4	IN3	Digital input 3
5	OUT0	Digital output 0
6	OUT1	Digital output 1
7	RL0	Right limit switch Axis 0 / MOTOR 0
8	LL0	Left limit switch Axis 0 / MOTOR 0
9	RL1	Right limit switch Axis 1 / MOTOR 1
10	LL1	Left limit switch Axis 1 / MOTOR 1
11	IO_SUP	24 VDC I/O Supply input
12	IO_RTN	24 VDC I/O Supply Return
13	AIN0+	Analog input 0 +
14	AIN0-	Analog input 0 -
15	AOUT+	Analog output +
16	AOUT-	Analog output -

4.6.1.1 J5 Connection Instructions

Use shielded cables with twisted pairs, a minimum gauge of 24 AWG and up to 100 meters in length.

1. The mating connector can be locked using the provided locking clip and screw, see [Control Supply Mating Connector and Locking Clips Kit](#).
2. Ensure that the analog input's signal range does not exceed the range of $\pm 10V$. Exceeding the range may cause abnormal behavior of the drive and affect its performance.
3. Connect a 24 VDC power supply between pins 11 and 12.

4.6.2 24 VDC I/O Power Supply Specifications

The 24 VDC power supply should comply with the following specifications:

Table 4-10. 24 VDC I/O Power Supply Specifications

	Description
Type	Isolated Low noise UL certified
Output voltage range	24 VDC $\pm$ 20%
Output current	>1A

Example: Lambda, P/N LS100-24

4.6.3 J10 PEG and Analog Input

4.6.3.1 Connector

- > Label: J10 PEG
- > Connector: 10-pin 2mm right-angle, by JST, PN S10B-PADKS-1
- > Mating Connector: by JST, PN PADP-10V-1-S
- > Pins: by JST, PN SPND-001T-C0.5

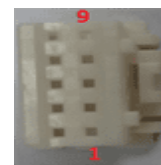


Figure 4-17. PEG Connector

Table 4-11. PEG Connector

Pin	Signal	Description
1	AIN1+	Analog input 1 +
2	AIN1-	Analog input 1 -
3	AGND	Analog ground
4	NC	Not connected
5	DGND	Digital ground
6	Shield	Shield
7	PEGO+	PEG 0 Output +
8	PEGO-	PEG 0 Output -
9	PEG1+	PEG 1 Output +
10	PEG1-	PEG 1 Output -

Pin ID on connector



4.6.4 Connection Instructions

Use shielded cables with twisted pairs, a minimum gauge of 24 AWG and up to 100 meters in length.

1. Ensure that the analog input's signal range does not exceed the range of $\pm 10V$. Exceeding the range may cause abnormal behavior of the drive and affect its performance.

4.6.4.1 Digital Inputs

All digital inputs (limit switch inputs, general purpose digital inputs, encoder registration mark inputs) are identical in structure and are factory configured to one of:

- > (D) or (N) Inputs & limits (and outputs) are 24V/SOURCE (PNP)
- > (S) Inputs & limits: 24V/SINK (NPN). (Outputs: 24V/SOURCE (PNP))
- > (R) Inputs & limits: 5V/SOURCE (PNP). (Outputs: 5V/SOURCE (PNP))
- > (T) Inputs & limits: 5V/SINK (NPN). (Outputs: 5V/SOURCE (PNP))
- > (U) Inputs: 24V/SOURCE (PNP). Limits: 24V/SINK (NPN). (Outputs: 24V/SOURCE (PNP))

The 4 digital inputs (IN0 to IN3) can be used as encoder registration mark inputs as well as general purpose inputs for any other need. For more information, see the *PEG and MARK Operations Application Notes*.

Table 4-12. Digital / MARK Inputs

Signal	Description
IN0	MARK0_1
IN1	MARK0_2
IN2	MARK1_1
IN3	MARK1_2

Unused limit switch inputs can also be used as general purpose inputs.

To use limits as general purpose inputs, clear the default response to the fault, using the following command:

```
FDEF(<axis number>).#RL = 0
```

or

```
FDEF(<axis number>).#LL = 0
```

To monitor the status of such inputs use the following command:

```
FAULT(<axis_number>).#RL or FAULT(<axis_number>).#LL
```

Example of an automatic routine that disables the motor when such an input is activated:

```
ON FAULT(0).#RL;  
  Disable (0);  
RET;
```

For more information, see the *SPiiPlus ACSPL+ Command & Variable Guide*.

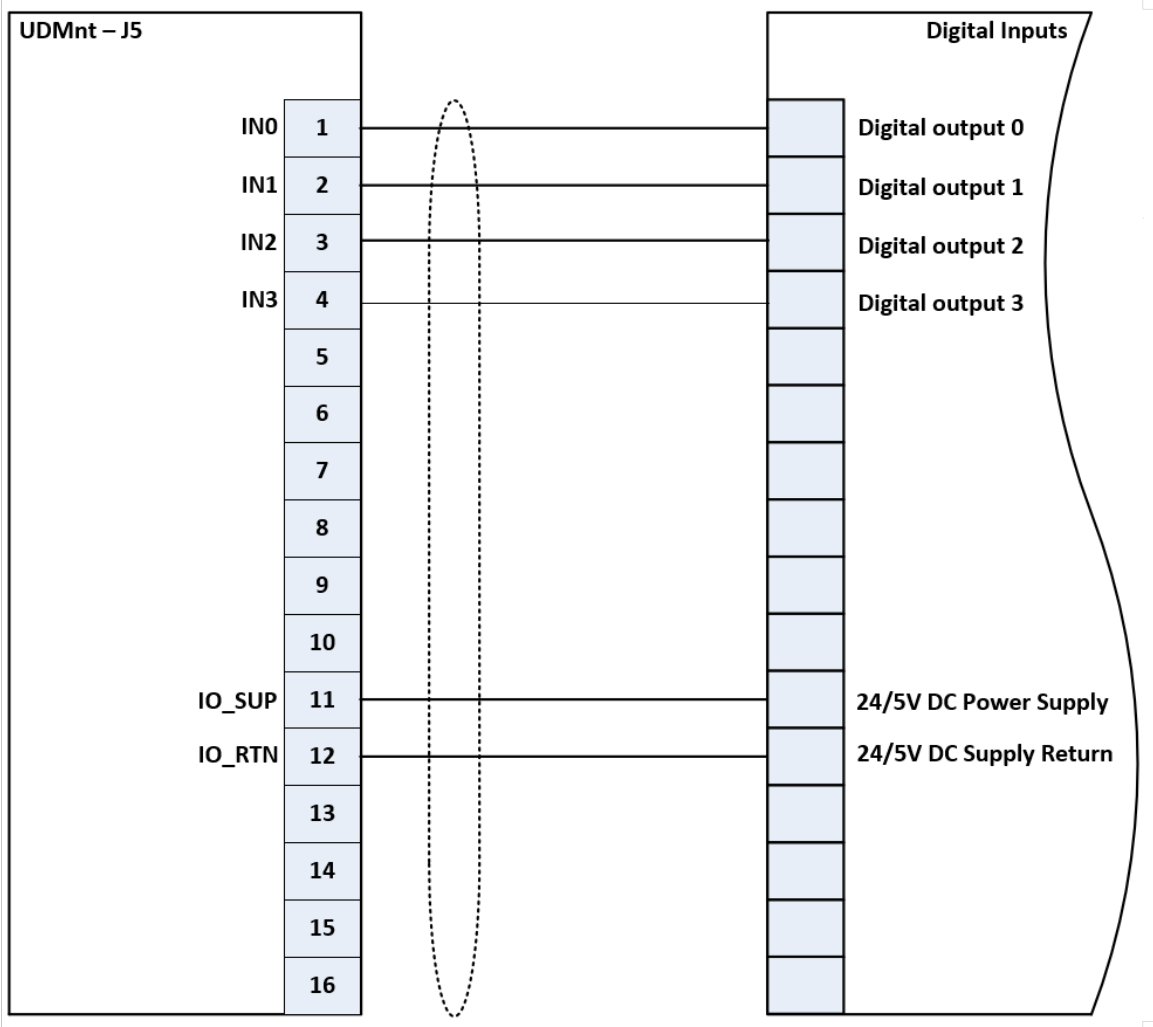


Figure 4-18. Digital Inputs Connection Diagram

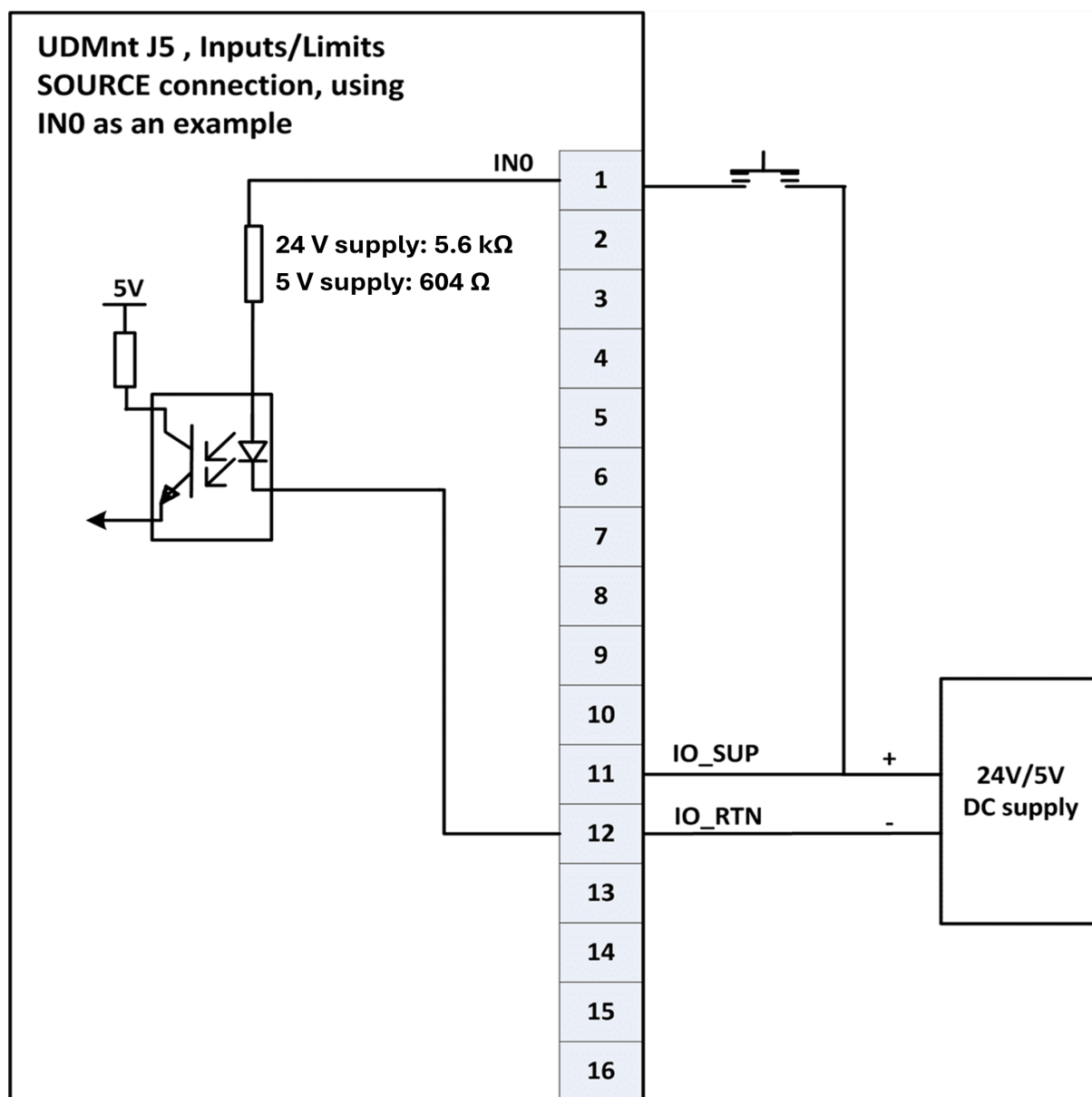


Figure 4-19. Example of Digital Inputs Source (PNP) Type Connection Diagram (Default)

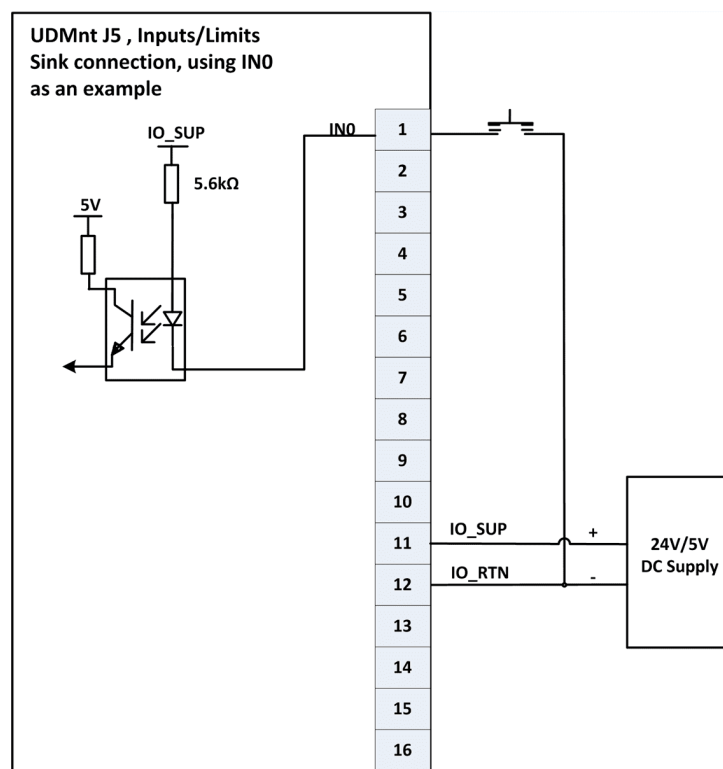


Figure 4-20. Example of Digital Inputs Sink (NPN) Type Connection Diagram

4.6.4.2 Digital Outputs

The 2 opto-isolated, 24 VDC, 0.1A digital outputs are factory configured as A Source (PNP) type.

The outputs can be used as motor brake drivers or as general purpose outputs. The default configuration is set as mechanical brake drivers.

Unused brake outputs can be used as general purpose outputs. Use the ACSPL+ command **SETCONF (29,<>,0)** as part of the application startup program to reconfigure the output. For more information see, the *SPIiPlus Command & Variable Guide*.

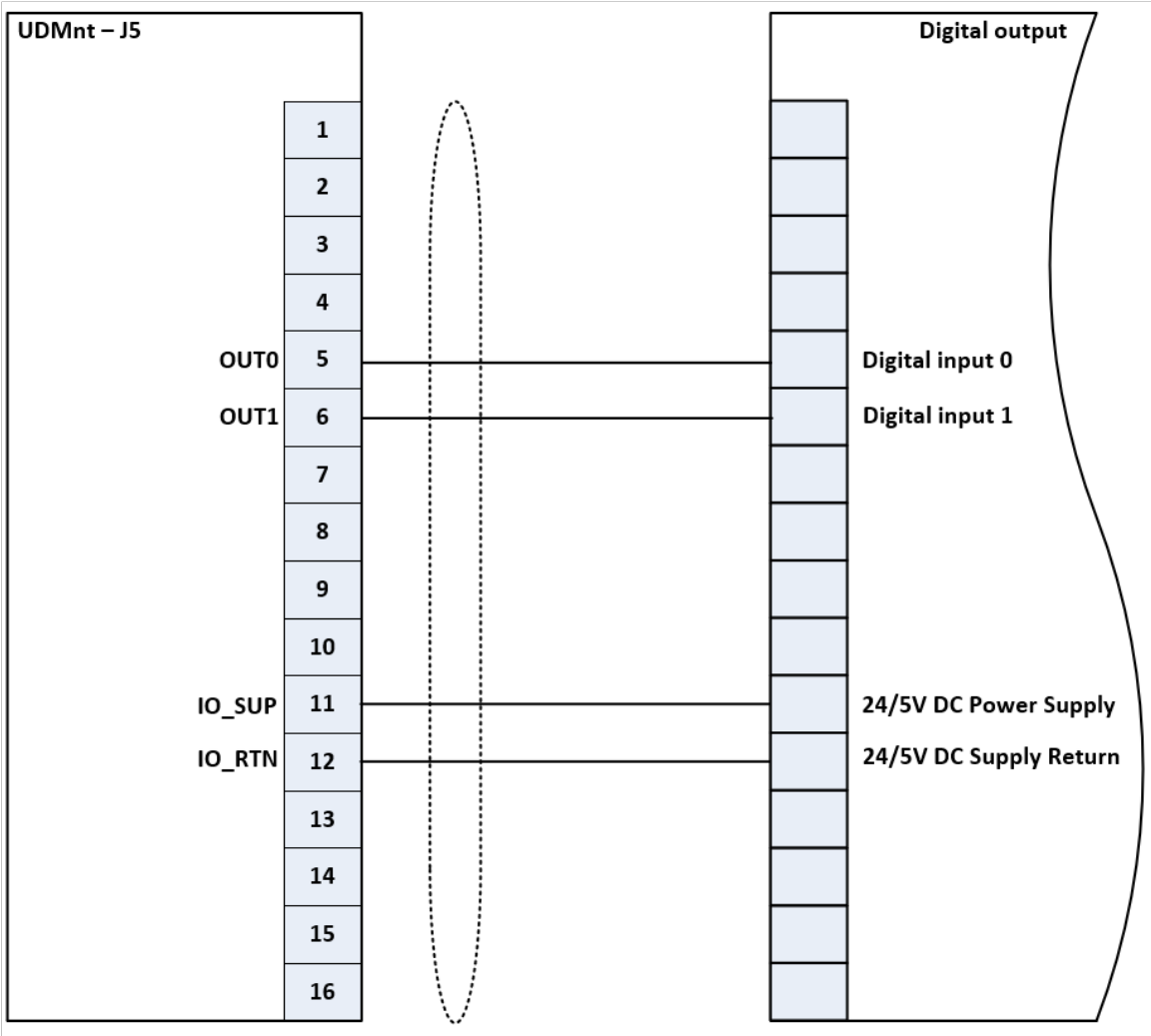


Figure 4-21. Digital Outputs Connection Diagram

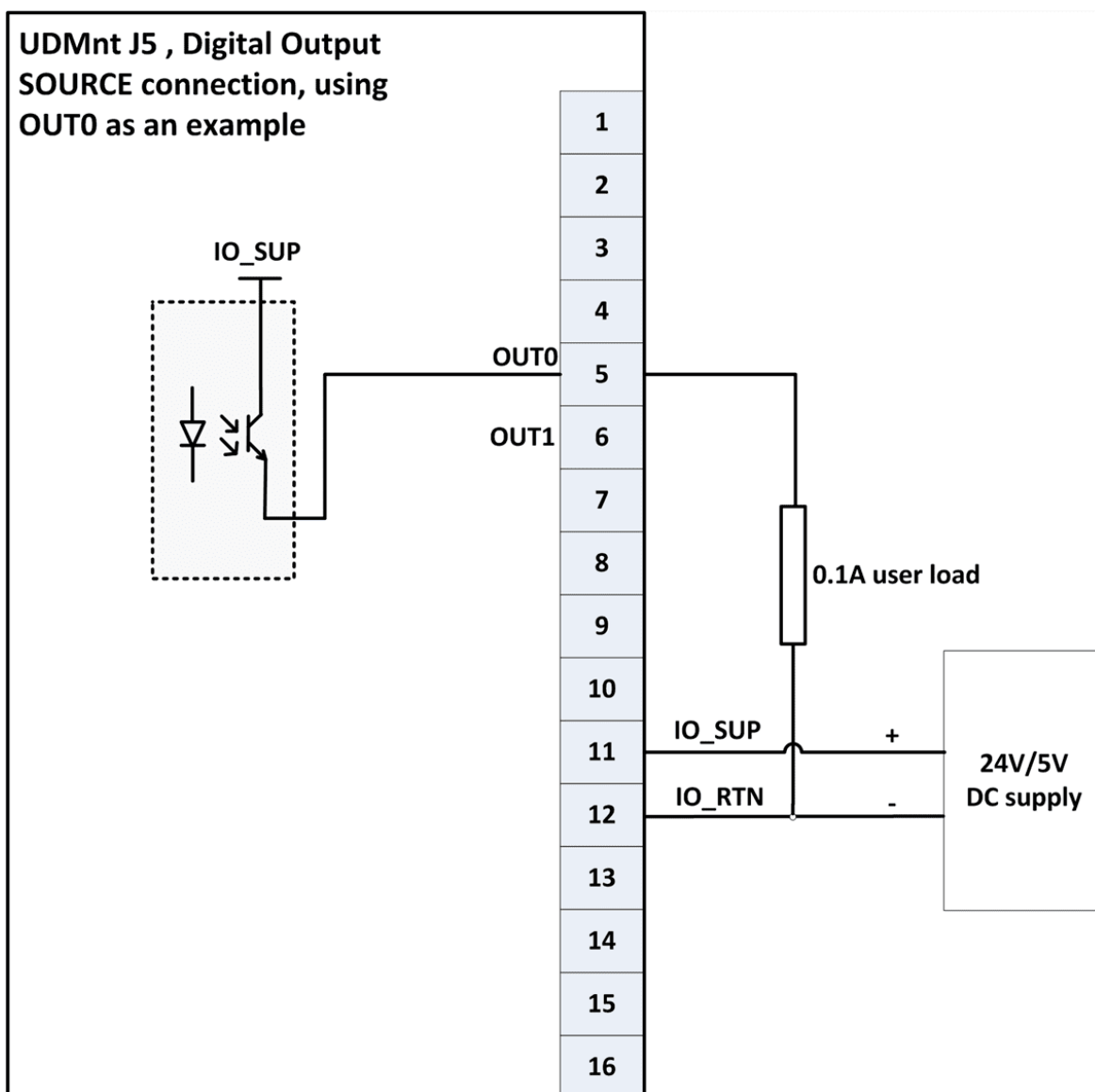


Figure 4-22. Example of Digital Outputs Source (PNP) Type Connection Diagram

4.6.4.3 Analog Inputs

The UDMnt has two (2) analog differential inputs, AIN0 and AIN1 with a range of +/-10V. The inputs are sampled by a 12-bit resolution A to D converter (ADC). AIN1 uses connector J10.

These inputs can also be used for feedback control purposes, which requires special software customized for the specific use.

Signal to Noise Ratio (SNR) = 56db AIN0 uses connector J5.

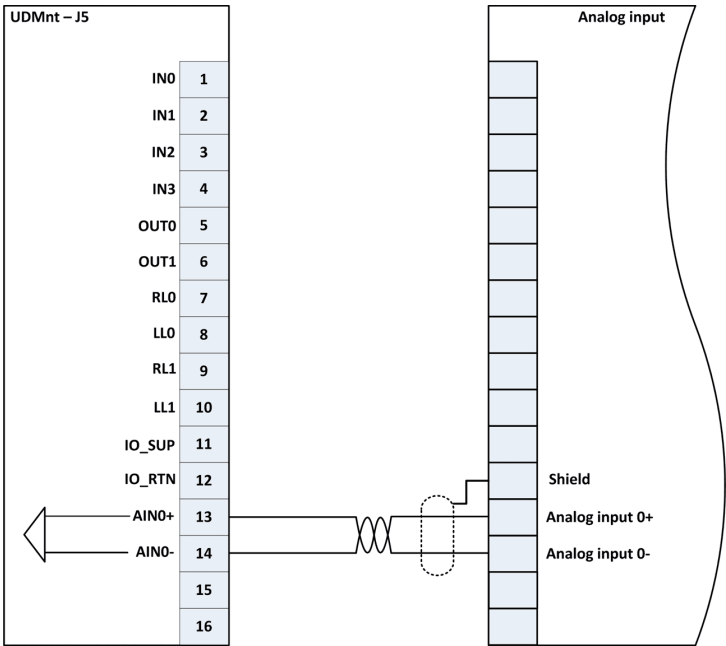


Figure 4-23. Analog Inputs Connection Diagram

4.6.4.4 Analog Output

The UDMnt has one analog differential output, AOUT +/-10V, with a range of +/-10V and with 10-bit resolution.



The user may choose to connect only one edge of the analog output, for single-ended connection, in such a case the negative edge (AOUT0-) should be left unconnected and the amplitude will be +/- 5V.

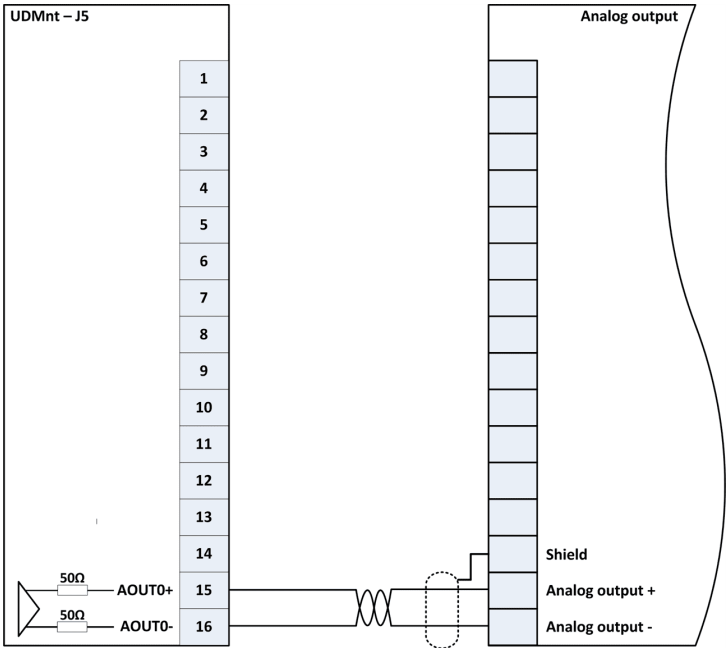


Figure 4-24. Analog Output Connection Diagram

4.6.4.5 Position Event Generation (PEG) Outputs

The UDMnt has two PEG_Pulse outputs. The user can program the controller to generate pulses at the exact location(s) of an axis, see *PEG and MARK Operations Application Note*.



The PEG operates only with incremental encoders.



PEG does not operate with absolute encoders.



With SinCos encoder, PEG the resolution is limited to four counts per encoder sine signal period (=encoder quadrature).

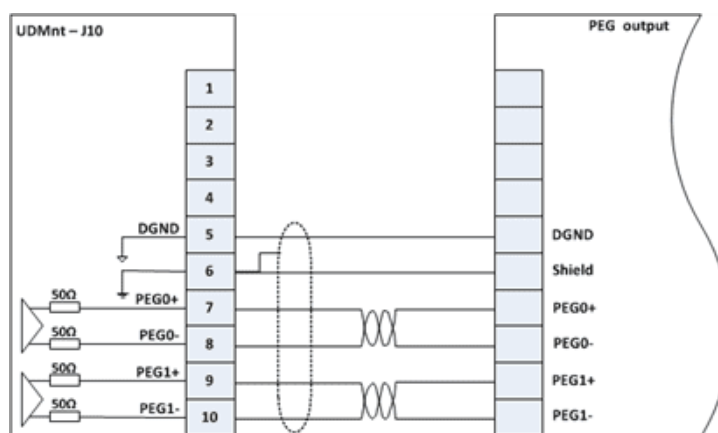


Figure 4-25. PEG Outputs Connection Diagram

To configure the PEG outputs, see the application note for PEG and MARK operations (*AN PEG and MARK Operations*).

5. EtherCAT

5.1 Connection Instructions

Use Ethernet cables CAT5e or better. ACS offers standard cables in different lengths, see "EtherCAT Cables" on page 11.

Connect the EtherCAT cable between J8 (ETHERCAT IN) and the EtherCAT output of the preceding EtherCAT master or slave.

1. Connecting J9 (ETHERCAT OUT)
 - a. When the UDMnt is not the last network node, connect the EtherCAT cable between J9 and EtherCAT IN of the next EtherCAT slave.
 - b. When the UDMnt is the last network node and a ring topology is used, connect J9 to the EtherCAT Master secondary port.
 - c. When the UDMnt is the last network node and a line topology is used, leave J9 unconnected.

5.2 J8 - ETHERCAT IN, J9 - ETHERCAT OUT

- > Labels: J8 ETHERCAT IN, J9 ETHERCAT OUT
- > Connectors: RJ45, 8P8C MJ Tab Up, by Wurth Elektronik, PN 615008137421
- > Mating cable with connectors: Standard Ethernet cables CAT5, see "EtherCAT Cables" on page 11.

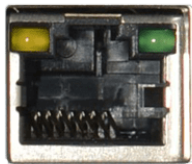


Figure 5-1. EtherCAT Connector

Table 5-1. EtherCAT Connector

Pin	Signal	Description
1	TD_0+	Transmit +
2	TD_0-	Transmit -
3	RD_0+	Receive +
4	NC	Not connected
5	NC	Not connected
6	RD_0-	Receive -
7	NC	Not connected
8	NC	Not connected

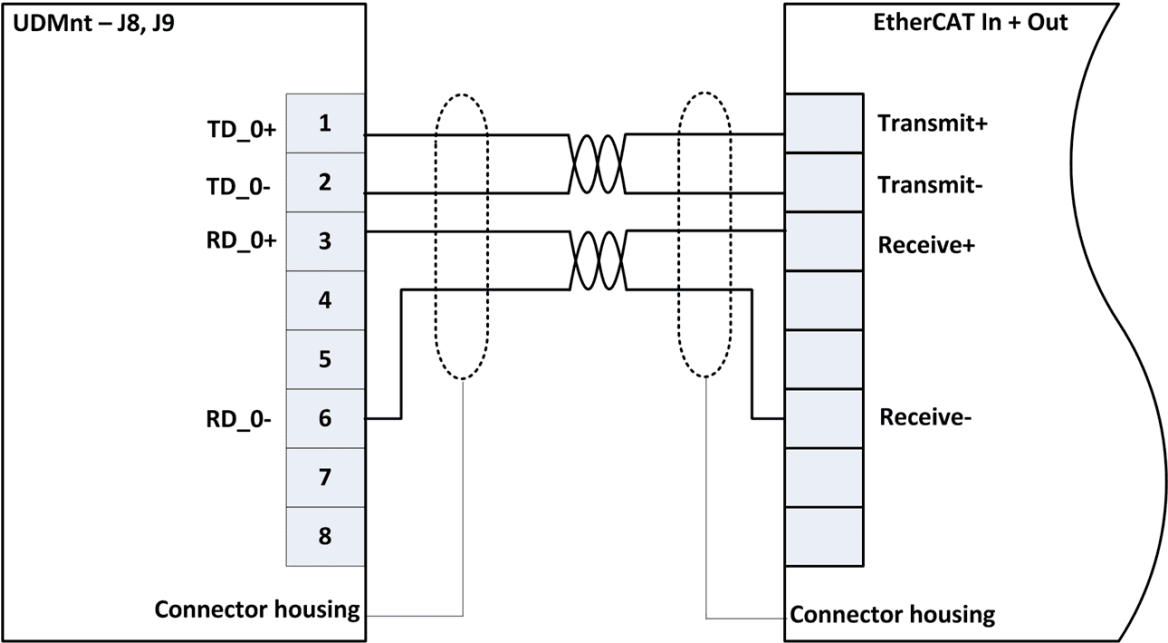


Figure 5-2. EtherCAT Connections

6. Product Specifications

This section provides detailed technical specifications of the UDMnt, including drive power ratings, interfaces, dimensions, environmental conditions, and standards.

6.1 Control

A standard comprehensive set of powerful algorithms to enhance accuracy, move & settle time, smooth velocity, stability and robustness.

- > Advanced PIV cascaded structure
- > Loop shaping filters
- > Gain Scheduling
- > Gantry MIMO control
- > Dual feedback / loop control
- > Disturbance rejection control

Optional ServoBoost algorithm (ordered with the controller) that provides better, more consistent servo performance, insensitive to noise and large changes in the system, contact ACS Support if you wish to apply the ServoBoost option.

See the following documents for more information about the various options:

- > *Motion Control Strategies to Obtain Better Performance*
- > *Controlling Gantry Tables with Cross Moving Axis*
- > *Application Note: Dual Axis PEG*
- > *ACSPL+ Programmer's Guide*

6.2 External Power Supplies Requirements

Control Supply, isolated

- > Input voltage 24 VDC $\pm$ 20%
- > Input current <1 A

Drive Supply, isolated

- > Input voltage 12 to 80 VDC
- > Not to exceed 83 VDC
- > Input current <8 A

6.3 Motor Drive

- > PWM, three- phase bridge
- > Switching method: advanced unipolar PWM
- > Digital current control with field oriented control and space vector modulation.
- > PWM switching frequency on the motor: 40 kHz
- > Current loop sampling rate: 20kHz
- > Programmable current loop bandwidth: up to 2kHz
- > Commutation: sinusoidal. Initiation with and without Hall sensors or commutation tracks

- > Protection: Overvoltage, motor phase-to-phase short circuit, motor phase-to-ground short circuit, overcurrent, overtemperature, and undervoltage of the following values:

Table 6-1. Motor Drive Specifications

Protection	Value
Motor phase short circuit	30A±5%
Overtemperature	100°C±3%
Overvoltage	85V±3
Undervoltage	9.5V±5%

Table 6-2. Motor Drive Specifications

Product	UDMnt...A...	UDMnt...B...	UDMnt...C...
Number of drives	1 or 2		
Motor types	Supports any of the following linear or rotary motors selected by the software settings: ▪ 2- or 3-phase Brushless Servo or Stepper ▪ DC Brush ▪ Voice Coil		
Input current [A]	<4.3	<8.6	<8.6
Motor phase current > Cont./Peak [A sine-amp] > Cont./Peak [Arms] > Peak current time [S]	2.5/5 1.8/3.6 1	5/10 3.6/7.2 1	10/20 7.2/14.4 1
Output power per drive @ full load, 80 VDC, Cont./Peak [W] Total	160/320 320/640	320/640 640/1280	640/1280 640/2560
Maximum output voltage on motor [VDC]	(Vin motor) x 92%		
Minimum Motor phase Inductance [μH] @ 80 VDC @ 48 VDC @24 VDC @12 VDC	50 30 15 8		

Product	UDMnt...A...	UDMnt...B...	UDMnt...C...
Maximum Heat dissipation per drive [W]	0.6	1.4	3.4



The two-phase motor has approximately a 30% de-rating factor on the Motor phase current and output voltage

6.4 Feedback Sensors Interfaces

There are two position feedback sensors interfaces, supporting the following types of sensors:

- > AqB,I and Clk/Dir (=Pulse/Dir) digital incremental encoders, with or without Hall sensors or commutation tracks.
 - > AqB,I Maximum input frequency: 12.5/50 million lines/ quad-counts/sec.
 - > Clk/Dir,I Maximum input frequency: 12.5 million clocks/sec.
 - > RS-422
- > AqB,I SinCos (analog) incremental encoder (with or without Hall sensors or commutation tracks). There are two types of SinCos encoder interfaces:
 - > 500kHz SinCos. This interface should be used with optical SinCos encoders.
 - > Maximum input frequency: 500kHz frequency.
 - > Maximum acceleration: 10^8 sine periods/second²
 - > 10MHz SinCos. This interface should be used with laser encoders for higher resolution while moving at high speeds.
 - > Maximum input frequency: 5MHz frequency
 - > Maximum acceleration: 10^8 sine periods/second²
 - > 1 Vptp, differential
 - > Programmable multiplication factor: x4 to x16,384
- > Absolute serial encoders of the following types:
 - > Data/Clock
 - > EnDAT 2.2
 - > BiSS-B/C (SSI)
 - > Data Line Only (non-return-to-zero NRZ)
 - > Tamagawa Smart-Abs
 - > Panasonic
- > Hall inputs:
 - > Two sets of three per axis.Type: single-ended, 5V, source, opto-isolated.
 - > Input current: <7mA.
 - > Maximum speed: 2kHz.

6.5 Digital Inputs / Outputs

- > Safety Inputs: Left and right limit inputs per axis:
 - > Single-ended, opto-isolated, Source (PNP, default) or Sink (NPN)
 - > On: connected to 24V or 5V
 - > Off: not connected
 - > Input current: 4-14mA
 - > Unused safety inputs can be used as general purpose inputs
- > Digital Inputs / Registration Mark:
 - > Quantity: Four
 - > Single-ended, Fast Opto-isolated, Source (PNP, default) or Sink (NPN)
 - > On: connected to 24V or 5V
 - > Off – not connected
 - > Input current: 4-14mA
- > Digital Outputs / Motor Brake Outputs:
 - > Quantity: Two
 - > Single-ended, Opto-isolated source (PNP, default) or Sink (NPN)
 - > 24V or 5V
 - > Output current: 100mA.
- > Position Compare Outputs (PEG):
 - > Quantity: Two
 - > RS-422
 - > Can be used as general purpose output
 - > Pulse width 26ns to 1.75ms



In some applications, the PEG output frequency may be limited to 33 MHz, and when an external SinCos multiplier is used it is limited to 25 MHz.

6.6 Analog Inputs

- > Quantity: Two
- > Differential, $\pm 10V \pm 5\%$,
- > Input impedance: $> 160\text{ k}\Omega$
- > ADC: 12 bit resolution. SNR = 56db

6.7 Analog Output

- > Quantity: One
- > Differential: $\pm 10V \pm 5\%$. Maximum output current: 1mA
- > DAC: 10bit resolution

6.8 Communication

Two EtherCAT ports, In and Out.

6.9 Dimensions

144 x 38.5 x 112.5 mm

(5.669 x 1.516 x 4.429 inch)

6.10 Environmental specification

- > Operational temperature: 0-50°C (32-122°F)
- > Storage and transportation range: -25 to +60°C (-13 to 140°F)
- > Humidity (operating range): 5% to 90% non-condensing

6.11 Compliance with Standards

6.11.1 Environment

RoHS

6.11.2 CE

- > Safety: IEC 61010-1:01
- > EMC: EN 61326:2002



CERTIFICATE OF CONFORMITY

With EN 61326-1: 2006, industrial locations equipment, Class A standard, harmonized under article 6(2) of EMC Directive 2004/108/EC

Certificate Number ACSEMC_EN.24782C

This certificate of conformity has been granted to the applicant based on the results of tests and evaluations, performed by Hermon Laboratories from August 12 to 14, 2013 on representative sample of the specified product.

Products description

Product family: UDMnt universal drive modules family
Tested item: Two axes universal drive module
Models: UDMnt2AXXXXXX
UDMnt2BXXXXXX
UDMnt2CXXXXXX
Tested item: One axes universal drive module
Models: UDMnt1AXXXXXX
UDMnt1BXXXXXX
UDMnt1CXXXXXX

Applicant/Manufacturer details

Name: ACS Motion Control Ltd.
Address: P.O.Box 5668, Ramat Gabriel Industrial Park, Migdal Ha'Emek
10500, Israel
Telephone number: +972 4654 6440
Fax number: +972 4654 6443

This is to certify that the tested product sample satisfies the requirements of the above listed standard/s.

Measurement/test results are contained in the test report: ACSEMC_EN.24782.

The comments in the associated (if applicable) test report/s shall be taken into account and used in conjunction with this certificate

Michael Nikishin,
EMC & Radio Group Manager
Hermon Laboratories Ltd.

September 1, 2013

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EXPERTS IN GLOBAL COMPLIANCE SOLUTIONS



EMC



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Product
Safety



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6.11.3 UL

- > UL Recognized UL508C





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